Figure 1: Unity Emacs search image

Figure 2: Metacity Emacs

```
# emerge --ask app-editors/emacs
```

On Arch, the Pacman software can help you in installing GNU Emacs:

```
$ sudo pacman -S emacs
```

Use the Zypper package manager in SUSE as shown below:

```
$ sudo zypper install emacs
```

Start-up

If you use the Unity interface, you can search for Emacs in the Dash, and it will show the Emacs icon that you can click to open the editor. This is illustrated in Figure 1.

On the Metacity interface or any other desktop environment with a panel, you can open GNU Emacs from *Applications* -> *Accessories* -> *Emacs*, as shown in Figure 2.

You can also open the editor from the terminal by simply typing 'emacs' and hitting the return key.

```
$ emacs
```

The version that I have used for this article is GNU Emacs 24.3.1.

Exit

To exit from GNU Emacs, you need to use *C-c C-q*, where 'C' stands for the Control key. You can also use your mouse and close the editor by clicking on the 'x', but GNU Emacs was designed to be completely usable with a keyboard, and I am going to encourage you to only use the keyboard shortcuts.

Concepts

While you can work using GUI editors, I am going to teach you to work entirely on the keyboard to experience the power of shortcuts in GNU Emacs. You can disconnect the mouse and your touchpad when working on GNU Emacs. Just as an exercise, I'd encourage you to remove your mouse completely and see how you can work with a computer for one day. You will realise that a lot of user interfaces are heavily dependent and designed for mouse interactions! By only using the keyboard, you can be blazingly fast and productive.

The keyboard shortcuts in GNU Emacs may seem to involve many keys. But please bear with me on this, because the way the shortcuts are designed, you will be able to remember them easily. As you practice, you will gain an insight into how consistently they have been defined.

GNU Emacs is a 'stateless' editor for the most part. By 'stateless', I mean that there are no specific state transitions that need to happen before you can use the commands. There does exist the concept of modes. When you open GNU Emacs, you will see menus, buffer and a mode line as illustrated in Figure 3.

As mentioned earlier, we will not be clicking on the menus or icons with a mouse, but only use keyboard shortcuts. Everything is a buffer in GNU Emacs. Each buffer can have one major mode and one or more minor modes. The mode determines the keyboard shortcuts that are applicable primarily on the buffer. Examples of major modes are given in Table 1.

Table 1

Mode	Description
Text mode	Writing text
HTML mode	Writing HTML
cc mode	Writing C, C++ and C-like programs
Dired mode	Handling files and directories
Shell mode	Working with shell
LaTeX mode	Formatting TeX and LaTeX files
Picture mode	Creating ASCII art
Outline mode	Writing outlines
SQL mode	Interacting with SQL databases
Lisp mode	Writing Lisp programs

The mode line exists below the buffer and it gives you a lot of information on the status of the buffer, such as what modes are active and other useful information. Below the mode line is the mini-buffer, where any commands you issue are indicated and prompts for user input are shown. This is the overall view of the default GNU Emacs interface.

In today's user interface applications, an application is treated as a window on a desktop. But, when you open GNU Emacs, you are actually opening a frame. A frame can be split into many windows. Everything is a buffer in GNU Emacs. So, you can

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